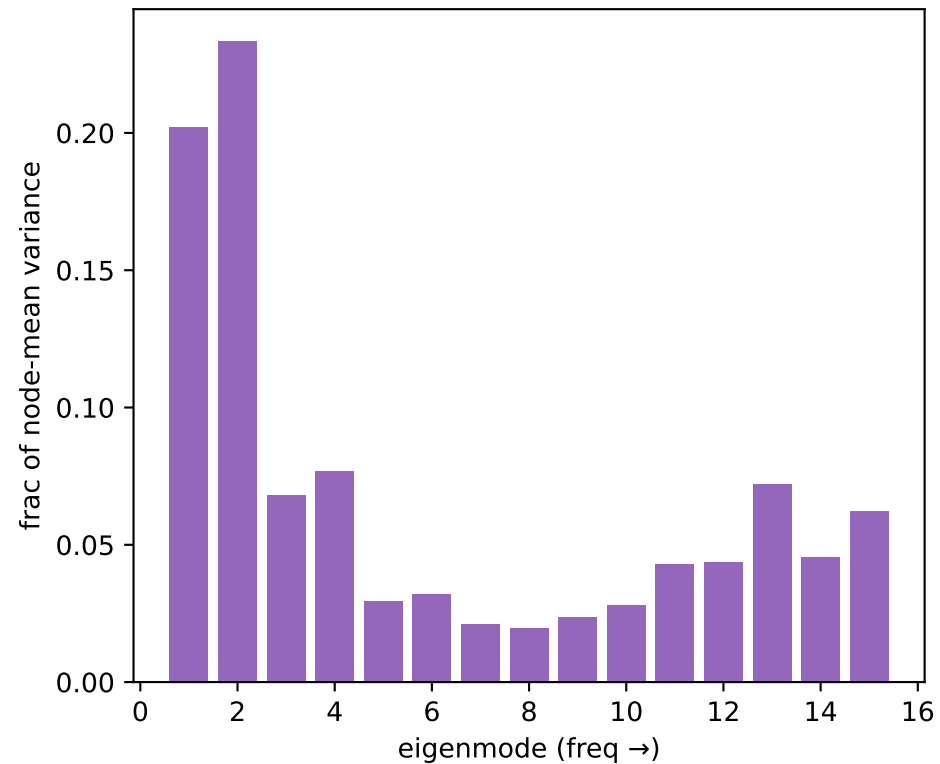
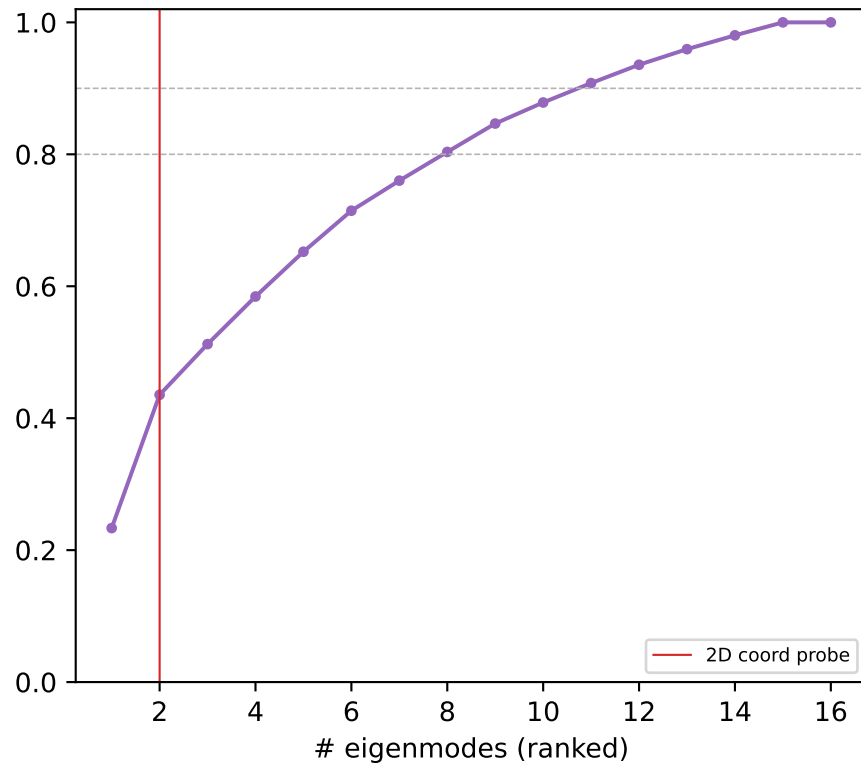


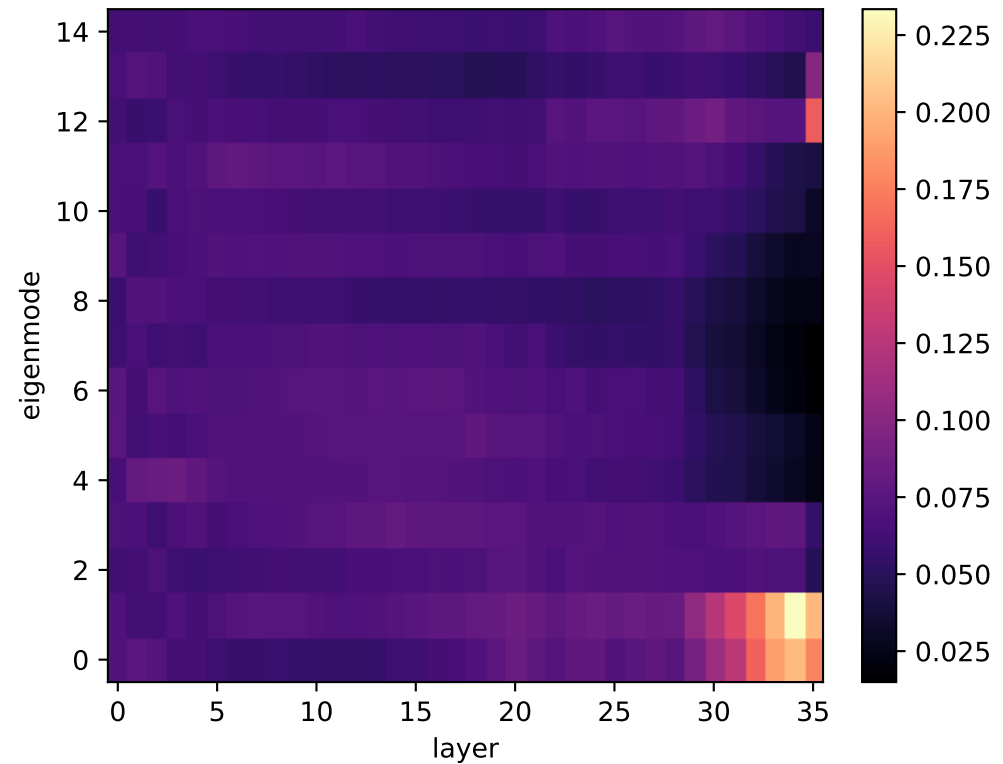
Qwen ring L34: power per Laplacian mode



cumulative variance vs #modes

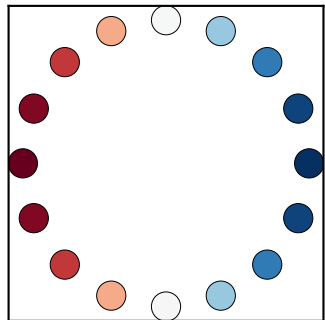


power: eigenmode x layer

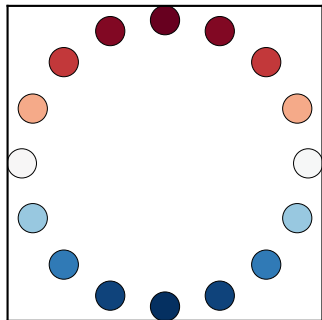


Qwen ring: Laplacian eigenmode DIVIDERS (% = variance share at L34)

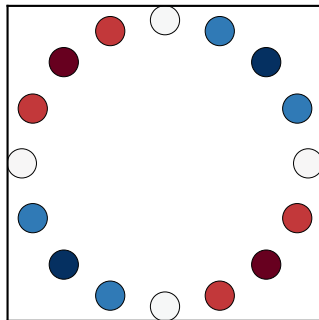
mode 1 $\lambda=0.15$ (20%)



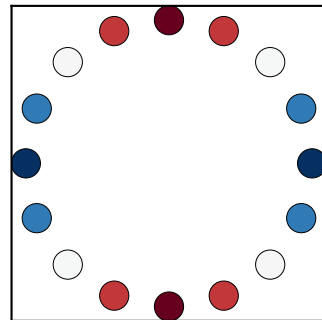
mode 2 $\lambda=0.15$ (23%)



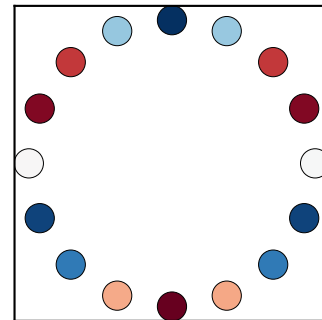
mode 3 $\lambda=0.59$ (7%)



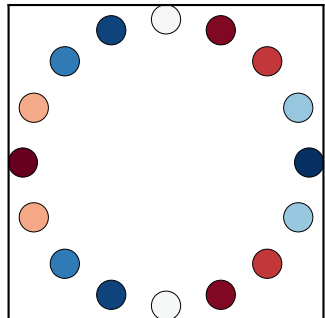
mode 4 $\lambda=0.59$ (8%)



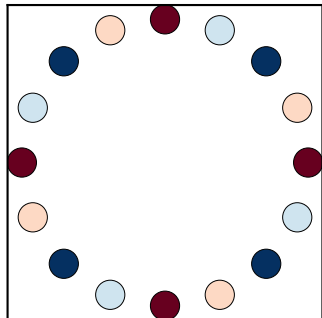
mode 5 $\lambda=1.23$ (3%)



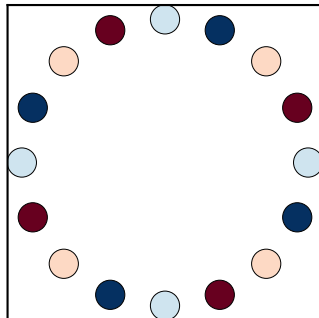
mode 6 $\lambda=1.23$ (3%)



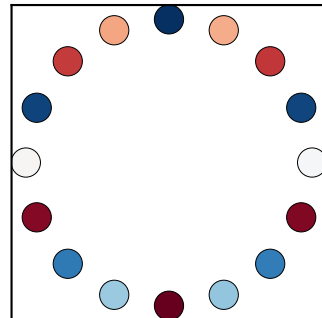
mode 7 $\lambda=2.00$ (2%)



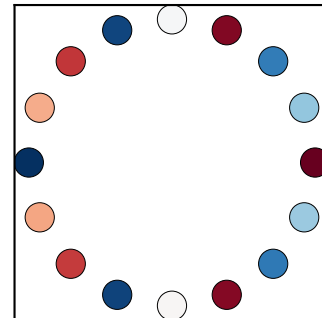
mode 8 $\lambda=2.00$ (2%)



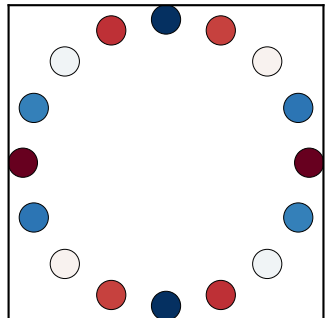
mode 9 $\lambda=2.77$ (2%)



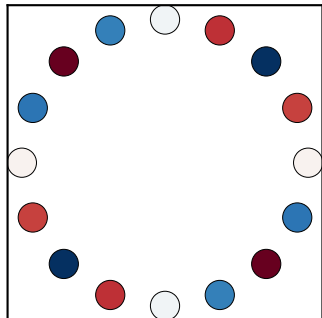
mode 10 $\lambda=2.77$ (3%)



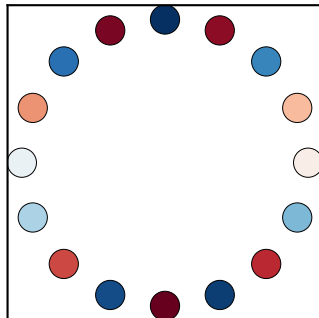
mode 11 $\lambda=3.41$ (4%)



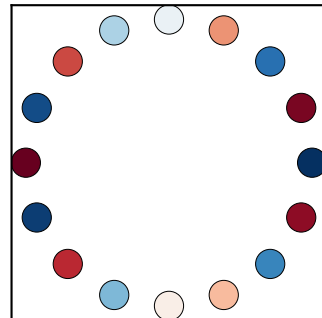
mode 12 $\lambda=3.41$ (4%)



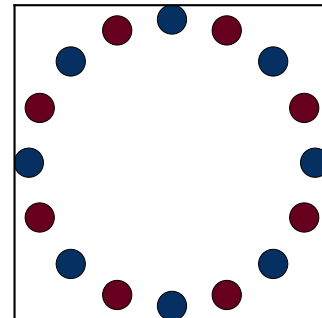
mode 13 $\lambda=3.85$ (7%)



mode 14 $\lambda=3.85$ (5%)

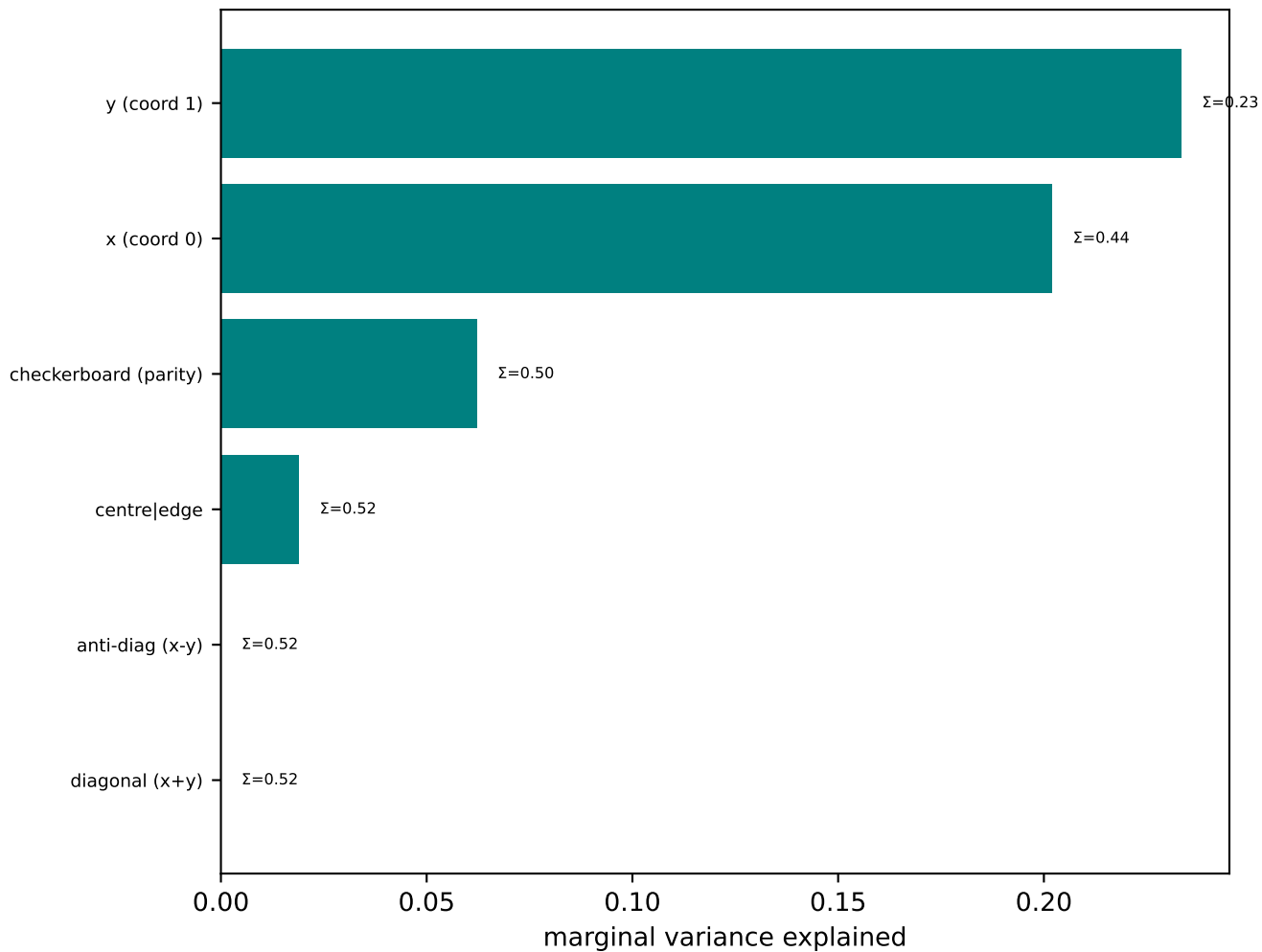


mode 15 $\lambda=4.00$ (6%)

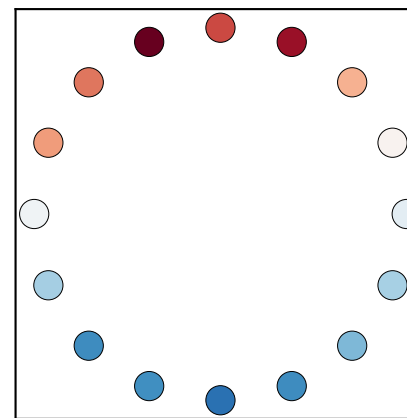


Qwen ring: named-cut basis (left) & model's own top axes (right)

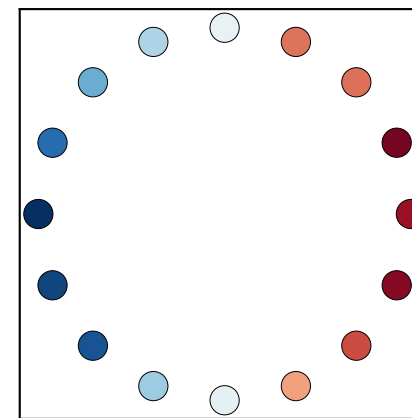
named cuts, greedy orthogonal (cumulative Σ)



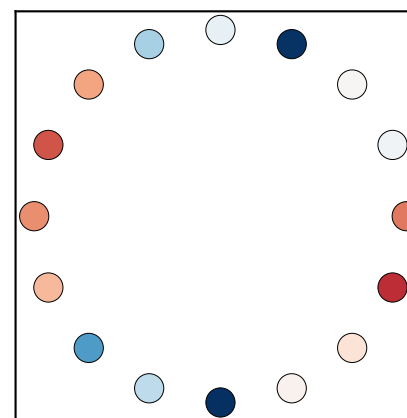
PC0 (24%) $\approx$ y (coord 1)



PC1 (20%) $\approx$ x (coord 0)



PC2 (8%) $\approx$ y (coord 1)



PC3 (7%) $\approx$ checkerboard (parity)

